MODULE III

Synchronous motor – Construction, Principle of operation, Effect of excitation on armature

Synchronous Motor - Features

- By construction it is very similar to synchronous generator
- It is a constant speed motor – speed decided by frequency
- Its power factor is controllable
- It requires both ac and dc supplies
  - AC for armature
  - DC for field excitation
- Synchronous motors are not self starting type
- Used for applications were constant speed operation is essential
  - Example: steel rolling mills, paper rolling mills

Construction of Synchronous motor

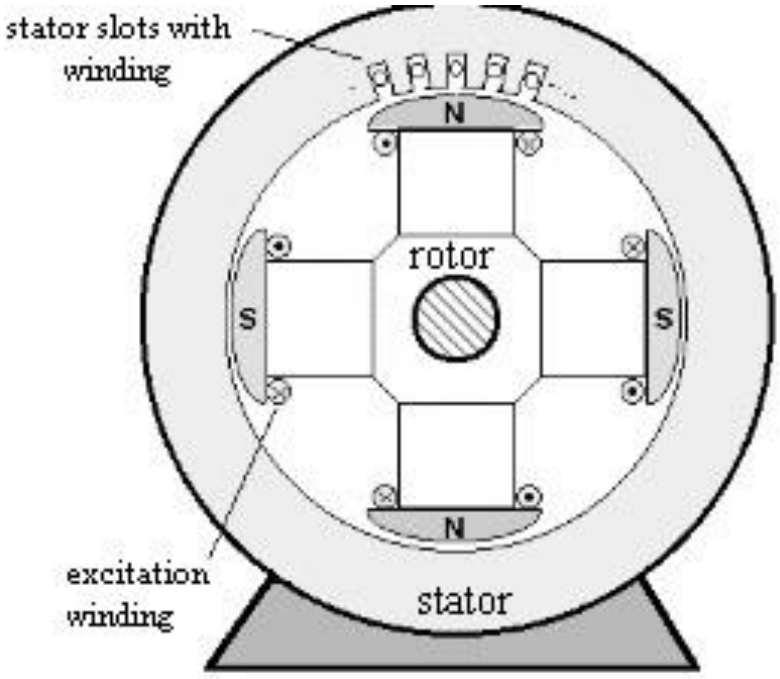
- It has mainly 2 parts
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1. Stator
2. Rotor

1. Stator or Armature

- Stator is the stationary part
- Main parts of **stator** are ***stator core, stator winding & stator frame***
- ***Stator core*** contain an iron ring made of silicon steel laminations
- Slots are cut on the inner periphery of the stator core, in which stator conductors are placed
- Stator core is laminated to reduce eddy current loss
- axial & radial ventilating ducts to provide efficient cooling
- Slots can be open, semi closed or totally closed
- Open slots are commonly used

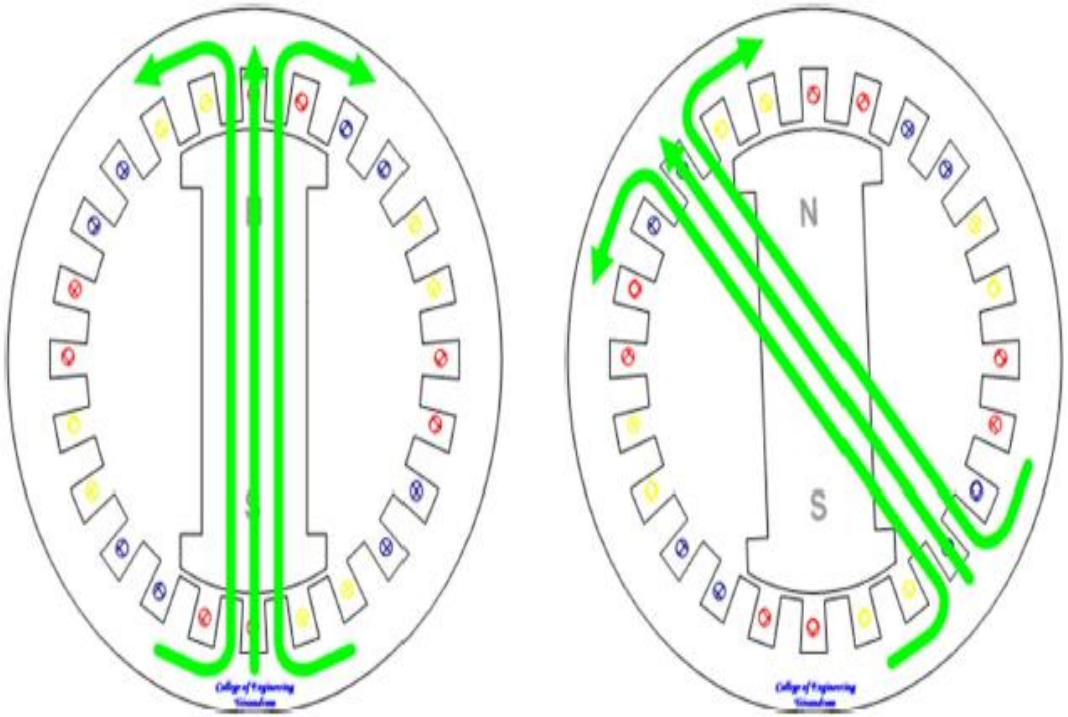
- Stator contain **a 3 phase winding** which is normally star connected
- Stator conductors are made of copper or aluminium
- Windings can be single layer or double layer, lap or wave, concentrated or distributed
- The whole structure is held in **a frame** made of cast iron or welded steel plate



2. Rotor or Field magnet system

- It is the rotating part of the machine
- Rotor contain Field magnets & are supplied with DC voltage
- For this external DC sources are used & exciting current is supplied to rotor through slip rings & brushes
- An exciter mounted on the motor shaft can also be used to provide necessary excitation.
- Normally, the rotor is having a salient pole structure
- Field poles contains laminated pole core & pole shoe (to reduce eddy current losses) and field windings

Why Synchronous Motors are not Self Starting ?

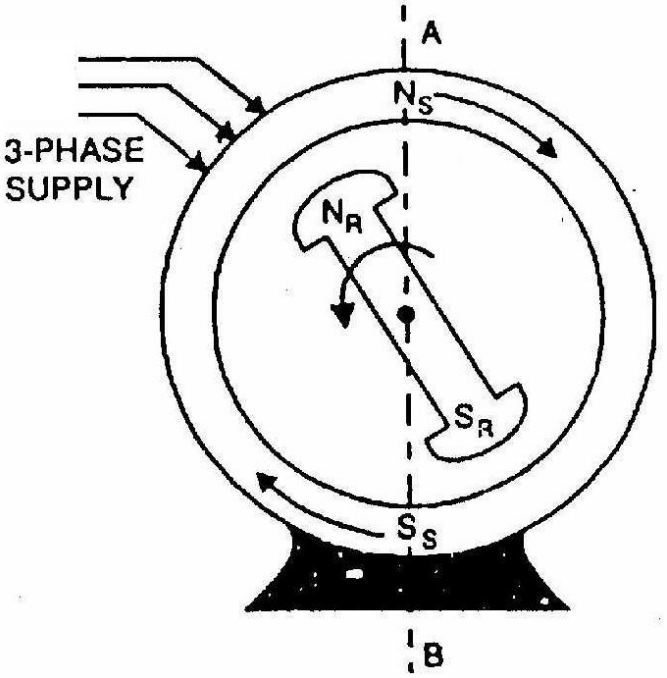


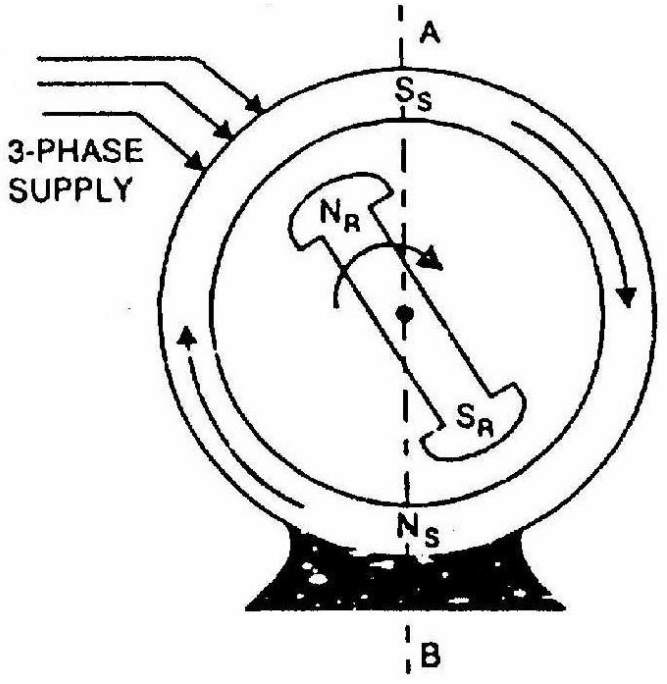
Due to the inertia of the rotor, it may not move quickly with the developed RMF when supply voltage is applied.

As the N and S poles of the RMF pass by, the rotor will tend to move in both clockwise and anticlockwise directions, but will not attain speed.

So synchronous motors are not self starting.

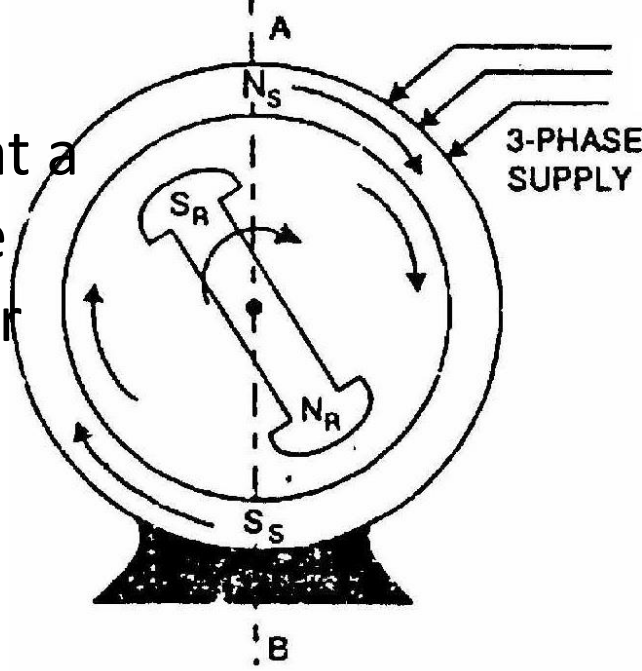
- Now stator North pole (Ns) & rotor North pole (Nr) will repel each other & rotor experiences a torque in anticlockwise direction
- After a period of half cycle, stator poles changes their position but rotor poles remain at previous position
- Now stator South pole (Ss) & rotor North pole (Nr) will attract each other & rotor experiences a torque in clockwise direction
- Since stator poles changes their polarity rapidly, rotor experiences torque in opposite directions in a cycle.
- Due to high inertia of rotor, motor fails to start

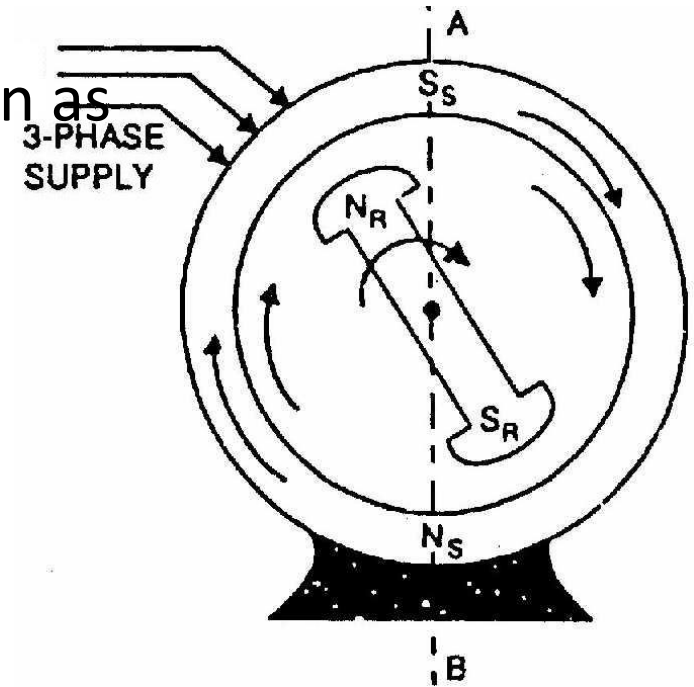




How to get unidirectional torque?

- If rotor is rotated by some external means at a speed near synchronous speed, in the same direction of stator magnetic field, then rotor will experience a unidirectional torque
- Now stator South pole (Ss) & rotor North pole (Nr) will attract each other & rotor experiences a torque in clockwise direction
- After a period of half cycle, stator & rotor magnetic field poles changes their position as shown in second figure
- rotor experiences a torque in clockwise direction.





- Under this condition, stator & rotor poles of opposite polarity experiences a strong attraction & the mutual attraction locks the rotor & stator together
- Now if the external prime mover is removed, rotor will continue to rotate at synchronous speed due to magnetic locking between stator & rotor

Starting methods of Synchronous motor

- Synchronous motor has no self starting torque & motor must first be brought upto synchronous speed by some external means for getting unidirectional torque in rotor.
- Various methods of starting of synchronous motors are

1. From DC source

- If a DC supply & a DC motor (shunt/compound) is available, the Synchronous motor is coupled to DC motor & is started using DC motor
- DC motor is coupled to Synchronous motor and dc motor is started & rotated at synchronous speed
- AC supply is given to Synchronous motor (make sure that rotating magnetic field & rotor of synchronous motor are rotating in same direction)
- Now DC field excitation is given & rotor gets locked to stator rotating magnetic field & continue to rotate at synchronous speed
- The DC motor is now disconnected from supply

2. By using an AC motor

- Here a small AC induction motor (called pony motor) is used for starting the Synchronous motor
- First the induction motor is started & will rotate Synchronous motor rotor at near synchronous speed
- AC supply is given to Synchronous motor (make sure that rotating magnetic field & rotor of synchronous motor are rotating in same direction)
- Now DC field excitation is given & rotor gets locked to stator rotating magnetic field & continue to rotate at synchronous speed
- The Induction motor is now disconnected from supply

3. By providing damper windings

- Synchronous motor can be made self starting by providing a special winding on rotor poles known as damper winding or squirrel cage winding
- Damper winding consist of short circuited copper bars placed in rotor poles
- When AC supply is given to stator, the synchronous motor will start as an Induction motor & rotates at a speed near synchronous speed
- Now DC field excitation is given & rotor gets locked to stator rotating magnetic field & continue to rotate at synchronous speed

Equivalent circuit of a Synchronous motor

- In a synchronous motor we are giving two supply, DC supply to rotor & 3 phase AC supply to stator
- When AC supply is given to stator, a current flows through stator windings
- Stator windings have resistance (R_a) & synchronous reactance (X_s)
- Also when a DC supply is given to rotor, a magnetic field is produced & it links stator conductors,
- As rotor rotates, flux linked with stator conductors changes & an EMF induces in stator windings. This is called back EMF (E_b) or counter EMF
- The back EMF opposes the stator supply voltage